



Society of Petroleum Engineers

SPE-227080-MS

From Voxels to Graphs an Efficient Approach to Estimate Elastic Properties of Digital Rocks

S. Alsinan, KAUST Upstream Research Center, Saudi Aramco, Thuwal, Saudi Arabia / King Abdullah University of Science and Technology / EXPEC Advanced Research Center, Saudi Aramco, Dhahran, Saudi Arabia;
K. Durdymyradov, King Abdullah University of Science and Technology; M. A. Al Ibrahim, EXPEC Advanced Research Center, Saudi Aramco, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227080-MS](https://doi.org/10.2118/227080-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Digital Rock Physics (DRP) enables computational simulation of elastic rock properties but remains limited by high computational cost and challenges in upscaling. To overcome these constraints, we propose a graph-based AI surrogate modeling framework that transforms 3D digital rock volumes into unstructured and weighted graphs, where nodes represent grains (or pores), and edges encode spatial connectivity. Using a dataset of 30,000 subsamples from five sandstone formations, we generate weighted graphs through a custom graph builder leveraging computer vision techniques. These graphs preserve the rock's complex geometry and topology more effectively than traditional voxel grids. A Graph Neural Network (GNN) is trained to predict bulk and shear moduli from these graph representations, achieving high accuracy on test data. The model also demonstrates strong generalization to new rock samples without additional training, outperforming conventional methods that require re-training. Moreover, the graph framework enables the upscaling of elastic property predictions from small sub volumes to full core plugs. This approach introduces a scalable, resolution-independent, and compact alternative to traditional DRP, addressing its main bottlenecks while preserving the fidelity of rock property predictions.

Introduction

Elastic properties, including P-wave velocity (V_p), S-wave velocity (V_s), and density, play a fundamental role in subsurface characterization, serving as key inputs for estimating petrophysical and geomechanical properties during exploration and field development. These properties are commonly inferred through seismic inversion techniques, with their accuracy influenced by the quality of seismic data, processing workflows, and the availability of well log data (Yilmaz, 2001; Tarantola, 2005). Alternatively, elastic properties can be measured directly through laboratory experiments, providing accurate and localized values under controlled conditions. The predictive range of such measurements can be extended through forward modeling using rock physics frameworks, which allow the evaluation of changes in elastic response due to variations in fluid saturation, mineralogy, cementation, and porosity (Mavko et al., 2020; Dvorkin

et al., 2014). Upscaling experimental data in geologically heterogeneous formations presents significant challenges. Accurately characterizing such systems requires an extensive and systematic sampling strategy to capture the full range of lithofacies and to quantify their elastic properties under diverse experimental conditions. This process is not only time-consuming but also logistically demanding. Moreover, it is often constrained by the limited availability of high-quality core samples and the need for specialized laboratory equipment capable of measuring elastic properties without altering the rock's intrinsic fabric. These limitations hinder the practical scalability of experimental approaches, particularly when attempting to represent the full variability present in natural reservoir systems. To address the limitations of traditional experimental approaches, synthetic rock analogs, either physically constructed in the laboratory or digitally synthesized, are increasingly being utilized (Husseiny and Vanorio, 2017; Andr  et al., 2013a; Suzuki et al., 2017). These analogs enable controlled studies in which individual variables affecting rock fabric and their impact on elastic properties can be systematically examined. Among these methods, Digital Rock Physics (DRP) has emerged as a particularly powerful tool, offering a computational framework for simulating elastic behavior under a wide range of conditions without the logistical and material constraints of physical experimentation (Andr  et al., 2013b). DRP integrates high-resolution 3D imaging with numerical modeling to generate virtual rock samples that faithfully represent the microstructural characteristics of real rocks. This non-destructive approach allows for precise manipulation of parameters such as mineralogy, pore structure, and fluid saturation, facilitating the exploration of scenarios that are impractical or impossible to replicate experimentally. Moreover, digital workflows enable rapid, parallel processing of large datasets, significantly enhancing the scalability and efficiency of rock property characterization. The strengths of experimental and digital approaches are most effectively realized when used in a complementary manner. Laboratory measurements provide critical ground-truth data that validate and calibrate DRP models, ensuring their physical accuracy and reliability. In turn, DRP offers the ability to interpret, generalize, and extrapolate from the limited and often sparse experimental datasets, enabling simulations across a broader range of conditions than can be feasibly tested in the lab. By integrating these methodologies, a more comprehensive and scalable understanding of the relationships between rock microstructure and macroscopic properties can be achieved. This combined approach enhances the predictive capability for key subsurface processes such as fluid flow, mechanical deformation, and reservoir performance.

While DRP has emerged as a powerful tool for characterizing the physical properties of geological formations, its broader adoption is limited by several critical bottlenecks. Notably, the considerable computational requirements of high-resolution numerical simulations constitute a major challenge (Saxena et al., 2018; Andr  et al., 2013b). The time required for individual simulations can range from several minutes to multiple hours, depending on the resolution and computational framework employed. This severely constrains scalability, particularly for upscaling applications, as each digital rock sample typically represents only a small portion of the full core volume (Saxena et al., 2019). Recent advances in machine learning, particularly deep learning, have demonstrated significant potential to accelerate numerical simulations through data-driven approaches (Wu et al., 2018; Hou and Cao, 2022; Liu and Mukerji, 2022; Liu et al., 2023; Ahmad et al., 2023; de Castro Vargas Fernandes et al., 2024). Convolutional Neural Networks (CNNs) are often the preferred choice due to their ability to efficiently exploit spatial feature maps while maintaining lower computational complexity compared to fully connected or transformer-based architectures. However, CNNs require extensive training data and often struggle to capture the complex, irregular spatial characteristics of rock fabric. They also lack flexibility with input size or resolution changes, making them sensitive to distribution shifts and limiting their scalability and generalization to unseen samples. To overcome these limitations, several studies have explored alternative network architectures better suited to capturing the complex spatial characteristics of digital rocks. These include point cloud-based representations (Kashefi and Mukerji, 2021) and hybrid models that combine CNNs with Graph Neural Networks (GNNs) (Alzahrani et al., 2023), aiming to improve structural representation while reducing computational overhead. More recently, Chung et al. (2024) demonstrated the use of GNNs as a

standalone approach for predicting the elastic moduli of digital rock samples. GNNs are particularly well-suited for modeling the unstructured, irregular geometries typical of porous media. Unlike CNNs, they do not require fixed input dimensions and can naturally operate on graph-based representations of the rock microstructure. Their ability to aggregate information across both local neighborhoods and global topologies enhances generalization performance across heterogeneous samples (Wu et al., 2021). Beyond digital rocks, GNNs have been widely adopted in surrogate modeling across materials science, chemistry, engineering and physics (Sanchez-Gonzalez et al., 2020; Wu et al., 2022; Reiser et al., 2022; Wang et al., 2021; Shoghi et al., 2024; Batatia et al., 2024), highlighting their potential for capturing complex physical relationships with high computational efficiency.

In this work, we present a technique that leverages 3D graph representations of digital rocks to accelerate the simulation and prediction of elastic rock properties. The method is scalable, compact, and resolution-independent, addressing key limitations of DRP, particularly at pore-scale resolutions. High-resolution 3D digital rock volumes are transformed into unstructured graphs that capture the geometric and topological relationships among grains and pores. A GNN-based surrogate model is trained to predict bulk and shear moduli from these graph representations, offering a computationally efficient alternative to conventional grid-based DRP simulations. Building on the work of (Chung et al., 2024), we introduce several enhancements to the graph construction process, enabling dense 3D connectivity within pore and grain graph representations, an essential requirement for high-fidelity numerical flow simulations. We further propose a new graph-based architecture specifically tailored to capture the intricate interactions within rock microstructures. To support improved predictive accuracy, we incorporate new geometric and topological features at both node and edge levels. We also investigate the effect of different segmentation strategies on graph quality and model performance, aiming to identify the most effective method for representing rock microstructures. Furthermore, we examine the predictive contribution of pore-only and grain-only graphs, comparing their performance to that of combined representations. Finally, we benchmark our results against existing models using previously unseen rock samples to rigorously assess generalization capabilities.

Methodology

Digital Rocks

This study uses publicly available high-resolution 3D digital images from five sandstone formations; Bentheimer, Berea, Leopard, Kirby, and CastleGate (Neumann et al., 2025), as shown in Figure 1. Each rock volume measures $1000 \times 1000 \times 1000$ voxels and was subsampled into non-overlapping $100 \times 100 \times 100$ voxel cubes. To expand the dataset, axial rotations along the x, y, and z axes were applied, resulting in 30,000 total subsamples. To evaluate generalization performance, CastleGate subcubes (6,000) were withheld entirely from training. The remaining data was divided into training (80%), validation (10%), and test (10%) sets. The elastic properties of the subcubes were computed under dry conditions using the ELAS3D simulator developed by NIST (Garboczi, 1998), which provides full stress and strain tensors for each sample. The resulting dataset spans a broad range of physical properties; bulk moduli from 5 to 35 GPa, shear moduli from 1 to 43 GPa, and porosity between 0.03 and 0.82, capturing significant heterogeneity in rock microstructure (Fig. 1).

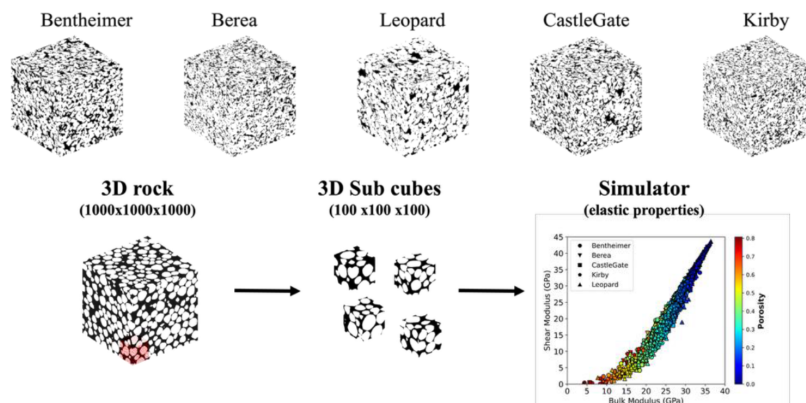


Figure 1—Figure captures the dataset used in this analysis (top) visualization of the 3D digital rocks, (bottom) data preparation workflow.

Graph Representation

We represent each 3D digital rock subcube as an undirected weighted graph, where nodes correspond to segmented pore or grain clusters and edges denote physical contact between them (Fig. 2). The segmentation is performed using the Watershed algorithm (Vincent and Soille, 1991), which effectively separates interconnected structures based on local intensity gradients (Fig. 2). This voxel-level segmentation provides a physically grounded graph topology.

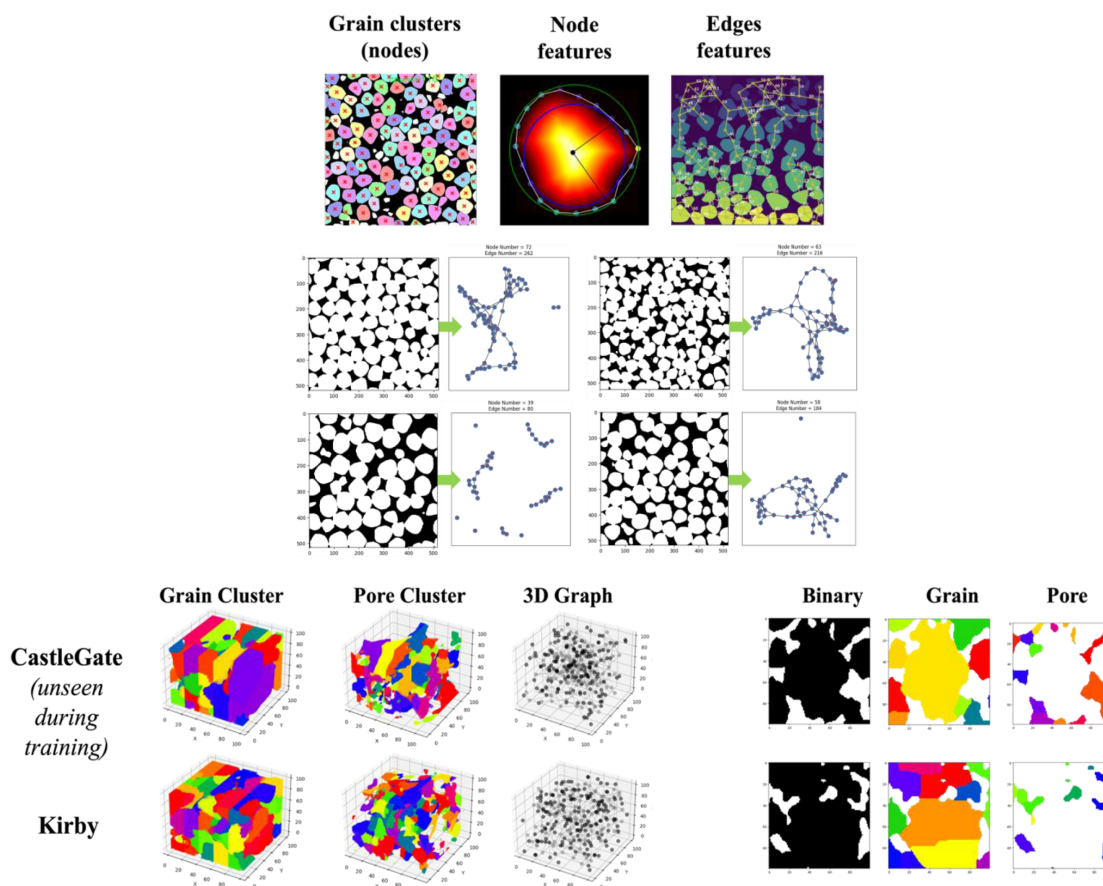


Figure 2—Graph representation of digital rocks. Top: Overview of graph features used to represent rock structure. Middle: Graph representations of random 2D slices from synthetic rocks, illustrating how variations in porosity and structure result in distinct connectivity patterns. Bottom: Graph representations of 3D subcubes from real rocks, showing pore and grain segmentation in both 2D and 3D, along with the resulting densely connected 3D graphs.

Unlike the Mapper algorithm (Chung et al., 2024), our graphs retain volumetric and contact-based information essential for mechanical interpretation. While (Chung et al., 2024) used Mapper-generated graphs based on topological simplifications, our approach encodes physical interactions directly from voxel-level segmentation. This results in a richer and more interpretable feature space, better suited for modeling elastic behavior (Fig. 3). Furthermore, our graph construction method is scalable, resolution-independent, and designed to preserve dense lateral and inter-layer connectivity.

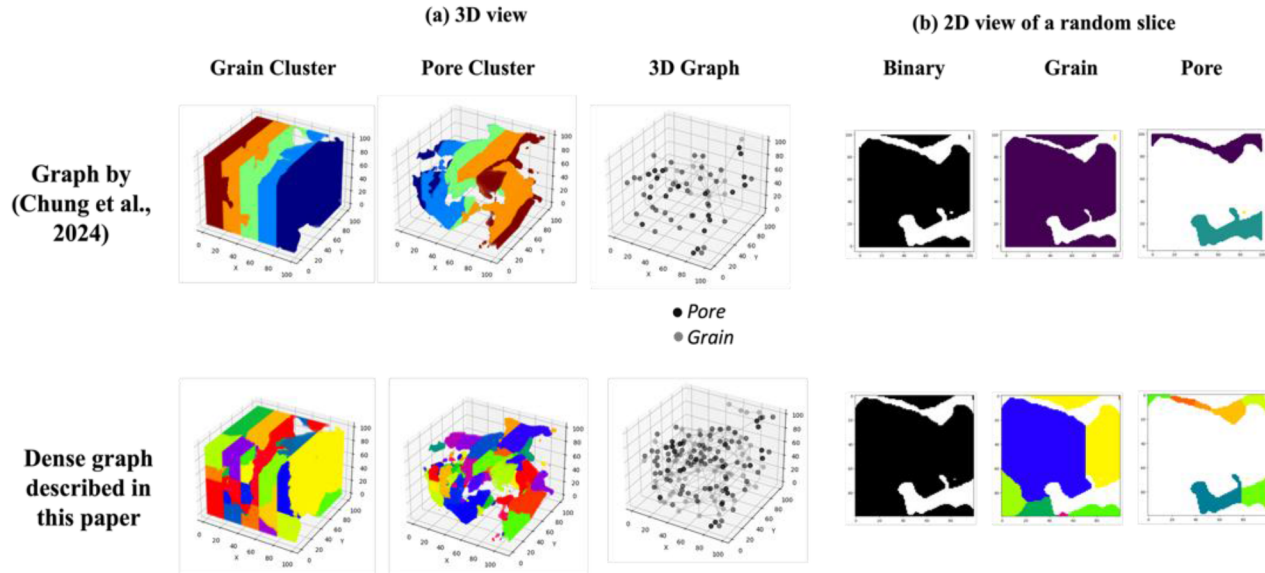


Figure 3—Comparing the dense graph representation in this paper to (Chung et al., 2024), (a) 3D view, (b) 2D view of a random slice.

Table 1 summarizes the graph features used for node and edge construction. Node features capture the geometry and morphology of individual pores or grains, incorporating shape descriptors such as area, perimeter, sphericity, roundness, major and minor axis lengths, as well as contact metrics like the number of interfacial pixels. Each node is located at the centroid of its corresponding cluster, providing a consistent spatial reference. Edge features quantify the connectivity between nodes using both the Euclidean distance between centroids and an area-weighted distance metric. In addition, edge weights reflect the number of interfacial voxels shared between connected clusters, providing a physically meaningful measure of mechanical connectivity. This feature design preserves both the spatial organization and mechanical relevance of the rock structure, enabling the resulting graph representation to support accurate, topology-aware modeling of elastic properties. The connection between clusters is assigned by a distance measure equal to or less than the sum of the radii of the two neighboring nodes. The adjacency matrix is stored in a compact (sparse) COOrdinate (COO) format, which reduces memory usage by approximately 95%. A similar graph construction approach is applied to pores, where pore geometry defines node embeddings, and edge weights are determined by pore throat sizes, capturing flow-relevant connectivity. This framework allows for a unified treatment of grain- and pore-scale features, making it applicable to a wide range of rock types, porosities, and imaging resolutions, and enabling flexible integration with machine learning models across scales.

Table 1—Graph features used for node and edge construction, capturing geometric, spatial, and connectivity properties of pores and grains.

Feature Type	Feature Name	Description
Node Features	Volume	Number of pixels in the cluster (grain or pore).
	Surface Area	Boundary area of the cluster.
	Sphericity	Measure of how close the shape is to a sphere.
	Roundness	Degree of shape sphericity.
	Axis Lengths	Lengths of the axes of the cluster
	Node degree	Number of neighbors of the node.
	Min Circumscribed Sphere	Radius of the smallest sphere enclosing the cluster.
	Max Inscribed Sphere	Radius of the largest sphere fully contained in the cluster.
	Number of Contact Pixels	Number of pixels in contact with neighboring clusters.
Node Location	Center of Cluster	Centroid position of the cluster, used as the spatial reference point for graph edges.
Edge Features	Distance Between Nodes	Euclidean distance between node centroids.
	Weighted Distance (Area-based)	Distance scaled by the average area of the connected nodes.

Graph Neural Network (GNN)

Graph Neural Networks (GNNs) are designed to operate on graph-structured data, making them ideal for irregular and non-Euclidean geometries. GNNs follow a message-passing paradigm, where each node aggregates information from its neighbors through a learnable function. This process allows the network to learn local structural patterns and propagate information across multiple layers, enabling the capture of both micro- and macro-scale dependencies. The aggregation of features at each layer ensures that the resulting node embeddings incorporate both geometric and topological context, crucial for predicting physical properties from digital rock graphs.

Similarly to [Chung et al. \(2024\)](#), in our implementation, we adopt Graph Isomorphism Network Convolution (GINConv) layers, selected for their strong expressive power in distinguishing graph structures. Unlike traditional GCN or GAT layers, GINConv utilizes a sum-based aggregation followed by a learnable multi-layer perceptron, which has been shown to be as powerful in terms of distinguishing graph structures. This property is particularly advantageous for modeling physical systems where fine-grained geometric distinctions influence macroscopic behavior.

[Fig. 4](#) illustrates the complete graph-to-property prediction pipeline. Our architecture includes two GINConv layers, each followed by Batch Normalization, ReLU activation, and a Dropout layer with a rate of 0.2 to prevent overfitting. Node features include geometric and spatial properties such as cluster volume, centroid coordinates, and principal axis lengths, while edge weights are derived from the number of interfacial voxels between connected clusters, encoding mechanical connectivity. After message passing, a global mean pooling operation condenses the learned node features into a fixed-size graph-level representation.

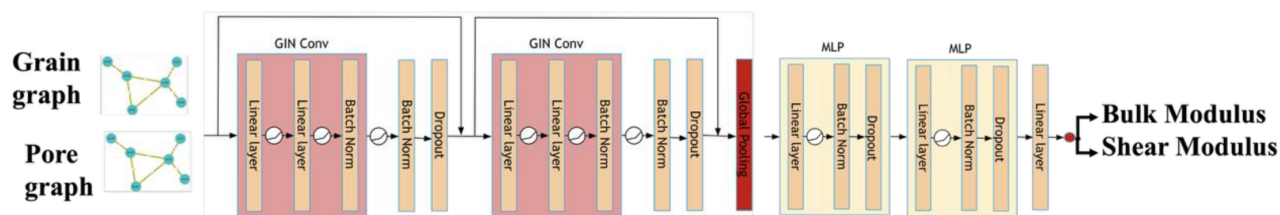


Figure 4—Graph Neural Network (GNN) architecture.

This pooled feature vector is fed into a Multi-Layer Perceptron (MLP) with three hidden layers. Each hidden layer is followed by ReLU activation and dropout. The final output layer has two units corresponding to the predicted bulk and shear moduli. The model is trained for 100 epochs using the Adam optimizer with a learning rate of 0.001 and optimized with the Mean Squared Error (MSE) loss function.

Results and Discussion

The Graph Neural Network (GNN) was trained using a supervised regression setup, with the ground truth bulk and shear moduli computed via ELAS:3D simulations. Across training runs, the model consistently converged within 30–50 epochs, indicating fast learning convergence due to the expressive power of graph-based representations. We observed stable training dynamics, with no significant overfitting, attributed to the high diversity of subsamples and the regularizing effect of the graph topology.

Fig. 5i shows the inference results on the test dataset for both bulk and shear modulus. Our GNN demonstrates strong predictive performance when trained using both pore and grain graph representations, achieving an R^2 of 0.84 for both bulk and shear modulus on the test data. An ablation study was conducted to evaluate the individual and combined contributions of pore and grain graphs (Fig. 6). Models trained using only pore graphs achieved R^2 scores of 0.81 (bulk) and 0.79 (shear), outperforming those trained on grain-only graphs (R^2 of 0.57 and 0.55, respectively). When both graph types were fused into a composite representation, predictive accuracy improved further, reaching R^2 of 0.84. These results emphasize that while grain information provides useful context, pore-scale connectivity plays a dominant role in controlling rock mechanical properties. The improved performance of the combined model demonstrates that each representation captures complementary aspects of rock structure, and their joint use is essential for robust prediction.

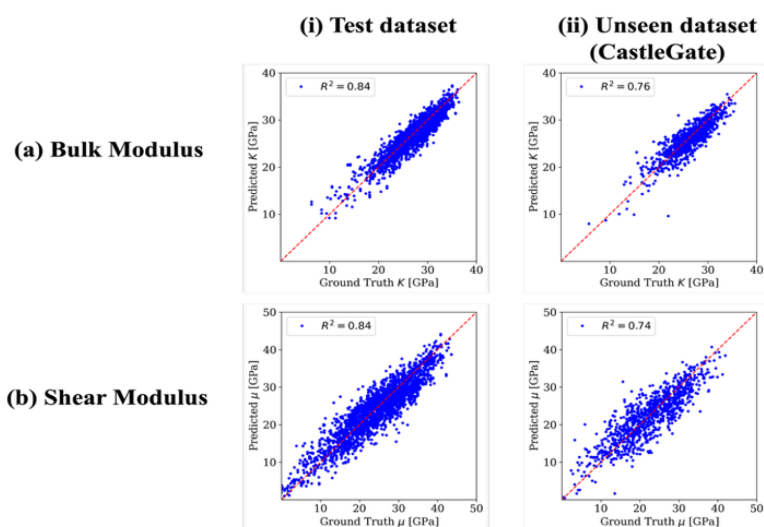


Figure 5—Predictions of the GNN trained with combined pore and grain graph representations on (i) the test dataset and (ii) the unseen CastleGate samples

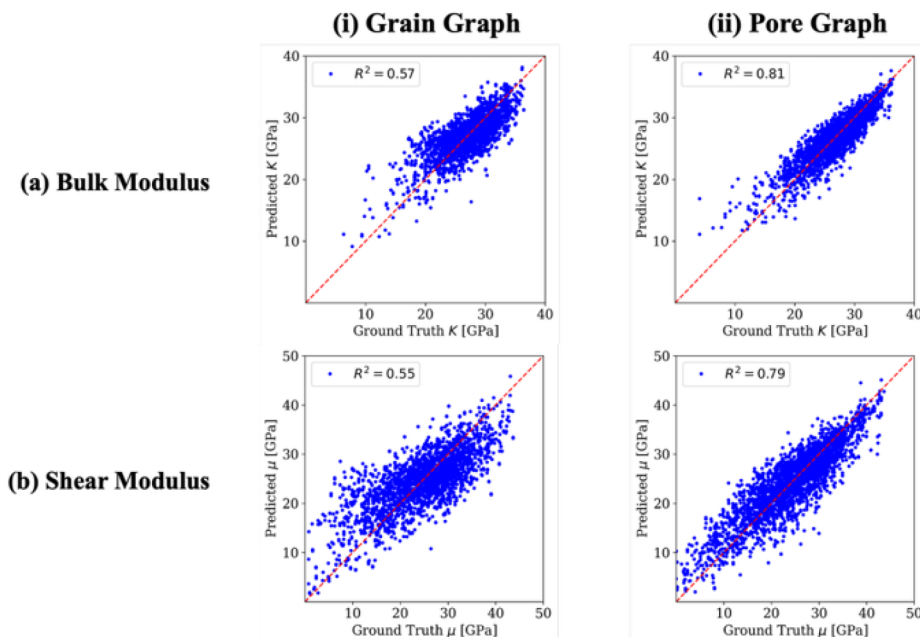


Figure 6—Prediction on test data using only (i) grain (ii) pore graphs for training.

To evaluate generalizability, we conducted out-of-distribution (OOD) testing by holding out entire rock formation (CastleGate) during training (Fig. 5-ii). The model maintained good predictive capability, achieving an average R^2 of 0.74 without additional fine-tuning. In contrast, conventional approaches, such as Convolutional Neural Networks (CNNs), typically require retraining on new data to achieve comparable accuracy. These findings highlight the model's ability to learn generalizable features from local graph structures, making it a promising candidate for deployment in scenarios where limited or no labeled data are available for new rock types. The flexibility of graph-based representations enables property upscaling, allowing predictions made at smaller scales to be aggregated to the full plug level. Since inference is independent of input size, the model seamlessly generalizes across scales.

Conclusion

We present an enhanced approach to representing structured 3D rock volumes as unstructured dense graphs, using a custom graph builder that captures detailed spatial relationships and connectivity within the rock. This improved representation retains complex topological features often lost in traditional grid-based models. The method enables scalable, resolution-independent prediction of rock properties while significantly reducing computational overhead. By addressing limitations in conventional digital rock physics, particularly in upscaling and simulation efficiency, it offers a compact and practical alternative.

Our Graph Neural Network achieves high predictive accuracy on test datasets and demonstrates strong and robust generalization to out-of-distribution samples from new rocks without requiring fine-tuning, making it well-suited for real-world digital rock applications. Future work will focus on extending this framework toward flow simulation, up scaling the results to the full core, and improving the underlying methodology to enhance predictive performance and broaden its applicability.

References

- Ahmad, Rasool, Mingliang Liu, Michael Ortiz, Tapan Mukerji, and Wei Cai. 2023. Homogenizing Elastic Properties of Large Digital Rock Images by Combining CNN with Hierarchical Homogenization Method. <https://arxiv.org/abs/2310.16802>

- Alzahrani, Mohammed K., Artur Shapoval, Zhixi Chen, and Sheikh S. Rahman. 2023. "Pore-GNN: A Graph Neural Network-Based Framework for Predicting Flow Properties of Porous Media from Micro-CT Images." *Advances in Geo-Energy Research* **10**(1): 39–55. <https://doi.org/10.46690/ager.2023.10.05>.
- Andrä, Heiko, Nicolas Combaret, Jack Dvorkin, Erik Glatt, Junehee Han, Matthias Kabel, Youngseuk Keehm, et al 2013a. "Digital Rock Physics Benchmarks—Part I: Imaging and Segmentation." *Computers & Geosciences* **50**: 25–32. <https://doi.org/10.1016/j.cageo.2012.09.005>.
- Andrä, Heiko, Nicolas Combaret, Jack Dvorkin, Erik Glatt, Junehee Han, Matthias Kabel, Youngseuk Keehm, et al 2013b. "Digital Rock Physics Benchmarks—Part II: Computing Effective Properties." *Computers & Geosciences* **50**: 33–43. <https://doi.org/10.1016/j.cageo.2012.09.008>.
- Batatia, Ilyes, Philipp Benner, Yuan Chiang, Alin M. Elena, Dávid P. Kovács, Janosh Riebesell, Xavier R. Advincula, et al 2024. "A Foundation Model for Atomistic Materials Chemistry." arXiv preprint arXiv:2401.00096.
- Chung, Jaehong, Waiching Sun, Wei Cai, and Tapan Mukerji. 2024. "Prediction of Effective Elastic Moduli of Rocks Using Graph Neural Networks." *Computer Methods in Applied Mechanics and Engineering* **421**: 116780. <https://doi.org/10.1016/j.cma.2024.116780>.
- Fernandes, J. de Castro Vargas, A. Duarte Vidal, L. Carvalho Medeiros, et al 2024. "Absolute Permeability Estimation from Microtomography Rock Images through Deep Learning Super-Resolution and Adversarial Fine Tuning." *Scientific Reports* **14**: 16704. <https://doi.org/10.1038/s41598-024-67367-1>.
- Dvorkin, Jack, Mario A. Gutierrez, and Dario Grana. 2014. *Seismic Reflections of Rock Properties*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511843655>.
- Garboczi, Edward J. 1998. Finite Element and Finite Difference Programs for Computing the Linear Electric and Elastic Properties of Digital Images of Random Materials. NISTIR 6269. National Institute of Standards and Technology.
- Hou, Zhiyu, and Danping Cao. 2022. "Estimating Elastic Parameters from Digital Rock Images Based on Multi-Task Learning with Multi-Gate Mixture-of-Experts." *Journal of Petroleum Science and Engineering* **213**: 110310. <https://doi.org/10.1016/j.petrol.2022.110310>.
- El Hussein, Ammar, and Tiziana Vanorio. 2017. "Porosity-Permeability Relationship in Dual-Porosity Carbonate Analogs." *Geophysics* **82**(1): MR65–MR74. <https://doi.org/10.1190/geo2015-0649.1>.
- Kashefi, Ali, and Tapan Mukerji. 2021. "Point-Cloud Deep Learning of Porous Media for Permeability Prediction." *Physics of Fluids* **33**(9). <https://doi.org/10.1063/5.0063904>.
- Liu, Mingliang, and Tapan Mukerji. 2022. "Digital Transformation in Rock Physics: Deep Learning and Data Fusion." *The Leading Edge* **41**(9): 591–598. <https://doi.org/10.1190/tle41090591.1>.
- Liu, Mingliang, Rasool Ahmad, Wei Cai, and Tapan Mukerji. 2023. "Hierarchical Homogenization with Deep-Learning-Based Surrogate Model for Rapid Estimation of Effective Permeability from Digital Rocks." *Journal of Geophysical Research: Solid Earth* **128**(2): e2022JB025378. <https://doi.org/10.1029/2022JB025378>.
- Gary Mavko, Tapan Mukerji, and Jack Dvorkin. 2020. *The rock physics handbook*. Cambridge university press.
- Neumann, Rodrigo, Mariane Andreeta, and Everton Lucas-Oliveira. 2025. "11 Sandstones: Raw, Filtered and Segmented Data." <https://www.doi.org/10.17612/f4h1-w124>.
- Reiser, Paul, Manuel Neubert, Andreas Eberhard, Oliver T. Unke, Kristof T. Schütt, Klaus-Robert Müller, Mark E. Tuckerman, O. Anatole von Lilienfeld, and Christian Kunkel. 2022. "Graph Neural Networks for Materials Science and Chemistry." *Communications Materials* **3**(1): 93. <https://doi.org/10.1038/s43246-022-00315-6>.
- Sanchez-Gonzalez, Alvaro, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. 2020. "Learning to Simulate Complex Physics with Graph Networks." In *Proceedings of the 37th International Conference on Machine Learning (ICML'20)*, Vol. **119**, 8459–8468. JMLR.org.
- Saxena, Nishank, Amie Hows, Ronny Hofmann, Faruk O. Alpak, Justin Freeman, Sander Hunter, and Matthias Appel. 2018. "Imaging and Computational Considerations for Image Computed Permeability: Operating Envelope of Digital Rock Physics." *Advances in Water Resources* **116**: 127–144. <https://doi.org/10.1016/j.advwatres.2018.04.001>.
- Saxena, Nishank, Amie Hows, Ronny Hofmann, Faruk O. Alpak, Jesse Dietderich, Matthias Appel, Justin Freeman, and Hilko De Jong. 2019. "Rock Properties from Micro-CT Images: Digital Rock Transforms for Resolution, Pore Volume, and Field of View." *Advances in Water Resources* **134**: 103419. <https://doi.org/10.1016/j.advwatres.2019.103419>.
- Shoghi, Nima, Adeesh Kolluru, John R. Kitchin, Zachary W. Ulissi, C. Lawrence Zitnick, and Brandon M. Wood. 2024. "From Molecules to Materials: Pre-Training Large Generalizable Models for Atomic Property Prediction." arXiv preprint arXiv:2310.16802. <https://arxiv.org/abs/2310.16802>.
- Suzuki, Anna, Noriaki Watanabe, Kewen Li, and Roland N. Horne. 2017. "Fracture Network Created by 3-D Printer and Its Validation Using CT Images." *Water Resources Research* **53**(7): 6330–6339. <https://doi.org/10.1002/2017WR021032>.
- Tarantola, Albert. 2005. *Inverse Problem Theory and Methods for Model Parameter Estimation*. Philadelphia, PA: Society for Industrial and Applied Mathematics. <https://doi.org/10.1137/1.9780898717921>.
- Vincent, L., and P. Soille. 1991. "Watersheds in Digital Spaces: An Efficient Algorithm Based on Immersion Simulations." *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(6): 583–598. <https://doi.org/10.1109/34.87344>.

- Wang, Haochen, Jianfa Han, Kai Zhang, Chuanjin Yao, Xiaopeng Ma, Liming Zhang, Yongfei Yang, Huaqing Zhang, and Jun Yao. 2021. "An Interpretable Interflow Simulated Graph Neural Network for Reservoir Connectivity Analysis." *SPE Journal* **26**(4): 1636–1651. <https://doi.org/10.2118/205024-PA>.
- Wu, Jinlong, Xiaolong Yin, and Heng Xiao. 2018. "Seeing Permeability from Images: Fast Prediction with Convolutional Neural Networks." *Science Bulletin* **63**(18): 1215–1222. <https://doi.org/10.1016/j.scib.2018.08.006>.
- Wu, Tailin, Qinchun Wang, Yinan Zhang, Rex Ying, Kaidi Cao, Rok Susic, Ridwan Jalali, Hassan Hamam, Marko Maucec, and Jure Leskovec. 2022. "Learning Large-Scale Subsurface Simulations with a Hybrid Graph Network Simulator." In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4184–4194. ACM. <https://doi.org/10.1145/3534678.3539045>.
- Wu, Zonghan, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, and Philip S. Yu. 2021. "A Comprehensive Survey on Graph Neural Networks." *IEEE Transactions on Neural Networks and Learning Systems* **32**(1): 4–24. <https://doi.org/10.1109/TNNLS.2020.2978386>.
- Yilmaz, Öz. 2001. *Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data*. 2nd ed. Tulsa, Oklahoma, U.S.A.: Society of Exploration Geophysicists.